MILITARY SPECIFICATION

DETONATOR, ELECTRIC, PATRIOT
PARTS, ASSEMBLING, LOADING AND PACKING FOR

This Specification is approved for use within U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the inspection requirements for the fabrication of parts, assembling, loading and packing for one type of one watt/one ampere Detonator, Electric, Patriot (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-148 - Aluminum Foil
MIL-P-387 - Pentaerythrite Tetranitrate (PETN)
MIL-L-46225 - Lead Azide, RD 1333
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FSC 1336
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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspections by Attributes
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-650	- Explosives, Sampling, Inspection and Testing
MIL-STD-810	- Environmental Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9279861	- Detonator, Electrical
9311342	- Fire Board Assembly (M143)
12630829	- Fire Board Assembly (M143E1)
8796522	- Wood Packing Boxes

2.1.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-190

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing on this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The materials, parts and explosives shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9279861 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Moisture content.

3.4.1 Lead azide. The moisture content of the lead azide complying with MIL-L-46225 at the loading station at the time of loading the detonators shall not exceed 0.3 percent.

3.4.2 Pentaerythrite tetranitrate (PETN). The moisture content of the PETN complying with MIL-P-387, Class 2 at the loading station at the time of loading the detonators shall not exceed 0.3 percent.

3.5 Plug.

3.5.1 Insulation resistance. The insulation resistance shall be 50 megaohms minimum at 1000 volts AC, root mean square (RMS) between:

a. The lead wires and the eyelet.
b. The shunted lead wires and the eyelet.

3.5.2 Pressure. The glass portion of the plug shall withstand a minimum dead load of 300 pounds without moving with respect to the eyelet. A plug shall be considered defective if either (1) the glass is forced from the eyelet during the test, or (2) the result of a visual examination reveals cracking or crazing of the glass.

3.6 Bridged plug soldering.

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3.6.1 Application. Solder shall be applied only to bright, clean surfaces. Fluxing and heating shall be such as to avoid flux inclusions. Operations shall be conducted to avoid insecure or pasty joints caused by vibration of the joint while the solder is setting. The bridge element shall be attached to the lead wires both electrically and mechanically in a sound manner.

3.6.2 Water soluble fluxes. The parts shall be rinsed and cleaned in hot water and then dipped in a solution containing five (5) ounces of sodium citrate (technical grade) per gallon of water. Following this, the parts shall be again rinsed in clean cold water and then in clean hot water and dried.

3.6.3 Water insoluble fluxes. Solvent cleaning to remove all visible trace of flux shall be employed. The solvent shall evaporate cleanly leaving neither solid nor liquid residue and shall not be capable of attacking the plug under normal conditions of usage.

3.7 Ignition assembly.

3.7.1 Charge holder assembly. When assembling the charge holder, care should be exercised to assure that neither the bridge material nor other parts of the sub-assembly are injured during the assembly operation. The charge holder shall be tightly assembled onto the plug with a seating force not less than eight (8) pounds. If the charge holder seating force is less than eight (8) pounds the sub-assembly shall be rejected.

3.8 Detonator, electrical, assembly.

3.8.1 Bridge resistance. The bridge resistance of the assembled detonator shall be not less than 1.0 ohms and shall be not greater than 1.5 ohms.

3.8.2 Insulation resistance. The insulation resistance between the shunted lead wires and cup shall not be less than 50 megaohms at 500 volts DC.

3.8.3 No fire. The detonator shall not fire when the bridge wire circuit is subjected to one ampere of current minimum, for not less than five (5) minutes.

3.8.4 Static discharge. The detonator shall not fire upon discharge of a 500 picofarad capacitor charged to 25,000 volts when a 5,000 ohm resistor is placed in series with test potential, the shunted bridge wire circuit and the case.

3.8.5 Functioning time. The detonator shall function in not greater than 100 microseconds when subjected to the discharge of the Patriot firing circuit in Figure 1 when charged with an input voltage of 28.0 ± 1.0 VDC.

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3.8.6 Output. The explosion of the Patriot detonator shall produce an indentation in steel having an average depth not less than 0.013 inch and no individual indentation value for the test sample shall be less than 0.012 inch. The test shall be performed in accordance with MIL-STD-331, Test 301.1, except that no sleeve shall be required.

3.8.7 Seal integrity. The detonator, coated with liquid leak detector, shall show no evidence of foam when subjected to a vacuum with a maximum absolute pressure of four (4) inches of mercury (Hg). The entire solder joint shall be observed for foam when the required vacuum has been obtained and held for a minimum of one minute.

3.8.8 Radiographic inspection. All detonator assemblies shall be radiographically inspected to assure that no air gap exists between PETN charge and ignition assembly, the proper location of all parts and that no parts are missing, malformed or damaged during assembly. Radiographic inspection shall be in accordance with MIL-STD-202, Method 209.

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PARTS LIST: See Fire Board Assembly Apparatus List Dwg. 9311342 (M143), 12630829 (M143E1), Sheet 2.

SCHEMATIC: See Fire Board Assembly, Dwg. 9311342 (M143), 12630829 (M143E1), Sheet 3.

	AMPLITUDE		DURATION	
	Minimum	Maximum	Minimum	Maximum
Charging Voltage (Note*)	27.0	29.0	10 Seconds	20 Seconds
Fire Pulse	20.0	28.0	100 Microsec	200 Microsec
Capacitor Voltage	80.0	100.0	3 Seconds	5 Seconds

*Note: Remove on or before Fire Pulse.

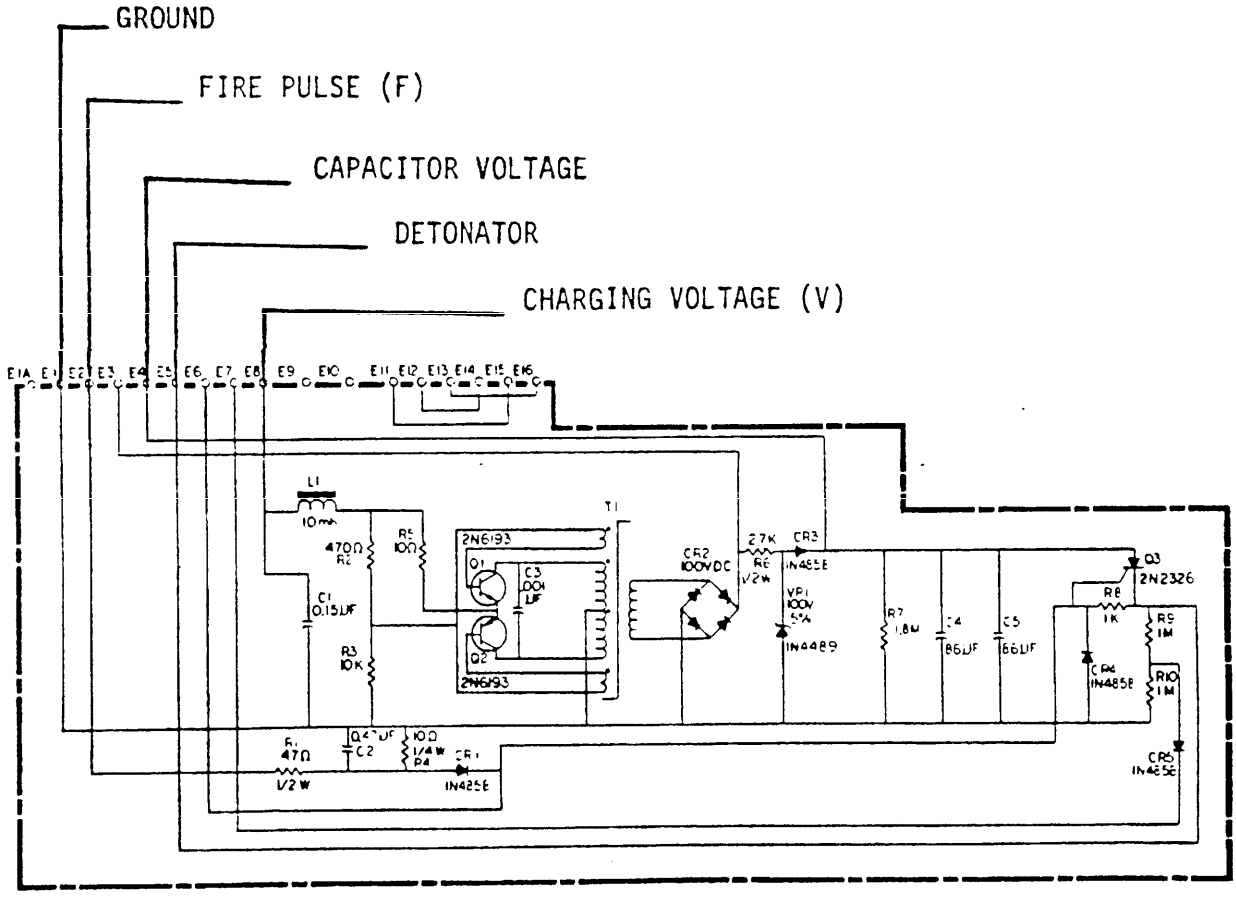


Figure 1.

Patriot Fire Circuit

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3.9 Environments.

3.9.1 Storage temperature. The detonator shall meet the function time, output requirements and electrical characteristics specified herein after being subjected to any temperature condition from -65°F ± 5°F (-53.0°C ± 2.8°C) to 160°F ± 5°F (71.1°C ± 2.8C°).

3.9.2 Operating temperature. The bare detonator shall meet the function time, output requirements and electrical characteristics specified herein at any temperature from -40°F -5°F (-40°C -2.8°C) to 150°F + 5°F (66°C + 2.8°C).

3.9.3 Transportation vibration (T.V). The detonator shall not function during the T.V. test. After the T.V. test, the detonator shall meet the function time, output requirement and electrical characteristics herein specified. The bare detonator shall be subjected to a sinusoidal vibration from 5.5 to 500 to 5.5 hertz (HZ) for a minimum of four (4) 15 minute Log sweeps, one hour total, along each of the three perpendicular axis at the following vibrator accelerations (see 4.5.12).

a. 1.0 inch double amplitude from 5.5 to 7 HZ.
b. Plus or minus (±) 2.5g peak from 7 to 38 HZ.
c. 0.036 inch double amplitude from 38 to 52 HZ.
d. Plus or minus (±) 5.0g peak from 52 to 500 HZ.

3.9.4 Flight vibration (F.V.). The detonator shall not function during the F.V. test. After the F.V. test, the detonator shall meet the function time, output requirement and electrical characteristics herein specified. The bare detonator shall be subjected to a random vibration along each of three principal axis for three (3) minutes minimum at the following levels (see 4.5.12).

a. 0.035 g^2/HZ from 100 to 2000 HZ.
b. Attenuated - 6 decibel (db)/octave from 50 to 100 HZ.
c. Attenuated - 6 decibel (db)/octave from 2000 to 3000 HZ.
d. TOTAL: 9.7 RMS g, 50 to 3000 HZ.

3.9.5 Temperature and humidity (T.H.). The detonator shall not function during the T.H. test. After the T.H. test, the detonator shall meet the function time, output requirement and electrical characteristics herein specified. The bare detonator shall be subjected to the Temperature and Humidity Test No. 105 of MIL-STD-331 with the exception that -80°F shall be changed to -65°F.

3.10 Test data. Test data from tests specified in 4.4.2.10 and 4.4.2.11 shall be submitted if requested by the procuring activity. Variable data shall be submitted for those requirements specified in 3.8.5 and 3.8.6. attribute data (go, no-go) may be submitted for all other requirements.

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3.11 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.12 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, unblended radii, cold solder joints, cracks, surface defects, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part of nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

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4.3.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Bridge Element	9279867	10
Plug, Prior to Glass Fill	9279866	10
Plug	9279866	10
Bridge Plug Assembly	9279865	10
Holder, Charge	9279864	10
Insulator	9279863	10
Ignition Assembly, Prior to Loading Lead Azide	9279862	10
Ignition Assembly	9279862	10
Cup	9279869	10
Detonator, Electrical, Patriot Assembly	9279861	100

4.3.1.1 Homogeneity. The first article shall comply with the homogeneity criteria of MIL-A-48078, 4.4.1. In addition, the first article shall contain:

a. Lead azide from one batch or blending of batches which results in homogeneity of all lead azide used in loading all the first article units.

b. PETN from one batch or blending of batches which results in homogeneity of all PETN used in loading all first article units.

c. Bridge element material from not more than one procurement lot.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.3.4	Detonator, Electric, Patriot Assemblies and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bridge Element (Dwg. 9279867) Examination for Defects	10		3.2	4.4.2.1
	Plug, Prior to Glass Fill (Dwg. 9279866) Examination for Defects	10		3.2	4.4.2.2
	Plug (Dwg. 9279866) Examination for Defects	10		3.2	4.4.2.3
	Bridge Plug Assembly (Dwg. 9279865) Examination for Defects	10		3.2	4.4.2.4
	Holder, Charge (Dwg. 9279864) Examination for Defects	10		3.2	4.4.2.5
	Insulator (Dwg. 9279863) Examination for Defects	10		3.2	4.4.2.6

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.3.4	Detonator, Electric, Patriot Assemblies and Components (continued)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ignition Assembly, Prior to Loading Lead Azide (Dwg. 9279862) Examination for Defects	10		3.2	4.4.2.7
	Ignition Assembly (Dwg. 9279862) Examination for Defects	10		3.2	4.4.2.8
	Cup (Dwg. 9279869) Examination for Defects	10		3.2	4.4.2.9
	Detonator, Electric, Patriot Assembly (Dwg. 9279861) Examination for Defects	100		3.2	4.4.2.10

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
4.3.4.1	Detonator, Electric, Patriot Assembly Tests				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Moisture Content of Explosives				
	Lead Azide (Dwg. 9279862)	4.4.3.1		3.4.1	4.5.1
	PETN (Dwg. 9279861)	4.4.3.1		3.4.2	4.5.2
	Environmental Tests				
	Group I at 70°F ± 10°F (Note a)	45		see 4.4.2.11	4.4.2.11
	Group I-A at 70°F ± 10°F (Note b)	15		see 4.4.2.11	4.4.2.11
	Group I-B at -40°F -5°F (Note b)	15		see 4.4.2.11	4.4.2.11
	Group I-C at 150°F +5°F (Note b)	15		see 4.4.2.11	4.4.2.11
	Group II Temperature and Humidity, 28 days (Note a)	15		see 4.4.2.11	4.4.2.11

NOTES: (a) Each sample unit shall be subjected to all the tests in the group.
(b) Samples for these sub-groups shall be selected from the initial Group I test samples. None of the detonators are to be subjected to more that one test of the subgroup.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Lead azide from one batch or blending of batches which results in homogeneity of all lead azide used in the loading of the entire lot of detonators.

b. PETN from one batch or blending of batches which results in homogeneity of all PETN used in loading the entire detonator lot.

c. Bridge element material from not more than two (2) procurement lots from one manufacturer.

4.4.2 Examinations. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test or Tables, sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279867
4.4.2.1	Bridge Element				NEXT HIGHER ASSEMBLY 9279865
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Center rib width		0.40%	3.2	Optical Comparator
102	Gap distance between rib and closest edge (2 places)		0.40%	3.2	Optical Comparator
103	Radius of rib (2 places)		0.40%	3.2	Optical Comparator
104	Tooth profile improper or missing		0.40%	3.2	Optical Comparator
105	Lead wire hole diameters (2 places)		0.40%	3.2	Optical Comparator
106	Outside diameter		0.40%	3.2	Optical Comparator
107	Lead wire holes symmetric to outside diameter		0.40	3.2	Optical Comparator
108	Lead wire holes symmetric to center rib		0.40%	3.2	Optical Comparator
109	Thickness		0.40%	3.2	Gage
110	Center rib width with respect to lead wire hole diameter		0.40%	3.2	Optical Comparator
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279866
4.4.2.2	Plug, Prior to Glass Fill				NEXT HIGHER ASSEMBLY 9279865
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter, max.		0.40%	3.2	Gage
102	Inside diameter concentric with outside diameter		0.40%	3.2	Gage
103	Outside diameter, body		0.40%	3.2	Gage
104	Length		0.40%	3.2	Gage
105	Surface finish		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279866
4.4.2.3	Plug				NEXT HIGHER ASSEMBLY 9279865
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of grooves, voids or scratches on glass surface between lead wires		100%	3.2	Visual
102	Insulation resistance between the leads		100%	3.5.1	4.5.2
103	Insulation resistance between shunted leads of eyelet		100%	3.5.1	4.5.2
104	Pressure test		0.40%	3.5.2	4.5.3
105	Contact prong and eyelets above flush to surface B		0.40%	3.2	Visual
106	Evidence of silver plate missing		0.40%	3.2	Visual
107	Evidence of cracks in rolled over surface		0.40%	3.2	Visual
108	Outside diameter, flange, min.		0.40%	3.2	Gage
109	Total length		0.40%	3.2	Gage
110	Perpendicularity		0.40%	3.2	Gage
111	Outside diameter, body		0.40%	3.2	Gage
112	Wire diameter, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279865
4.4.2.4	Bridge Plug Assembly				NEXT HIGHER ASSEMBLY 9279862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height of solder		0.40%	3.2	Gage
102	Evidence of flux after cleaning		0.40%	3.6.2 & 3.6.3	Visual
103	Excessive solder which obscures configuration		0.40%	3.2	Visual
104	Bridge element lifted after soldering		0.40%	3.6.1	Visual
105	Cold solder connection		0.40%	3.10	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Holder, Charge	1 OF 1	9279864
			NEXT HIGHER ASSEMBLY: 9279862

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Small inside diameter		0.40%	3.2	Gage
103	Thickness through bottom of counterbore		0.40%	3.2	Gage
104	Inside diameter concentric with outside diameter		0.40%	3.2	Gage
105	Outside diameter, min.		0.40%	3.2	Gage
106	Large inside diameter concentric to outside diameter		0.40%	3.2	Gage
107	Large inside diameter, max.		0.40%	3.2	Gage
108	Outside chamfer missing		0.40%	3.2	Visual
109	Radius on any fillets or corners missing		0.40%	3.2	Visual
110	Finish improper		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279863
4.4.2.6	Insulator				NEXT HIGHER ASSEMBLY 9279862
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, min.		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Inside diameter concentric to outside diameter		0.40%	3.2	Gage
104	Thickness, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279862
4.4.2.7	Ignition Assembly, Prior to Loading Lead Azide				NEXT HIGHER ASSEMBLY 9279861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bridge element not in proper position		0.40%	3.2	Visual
102	Evidence of foreign material on bridge element		0.40%	3.2	Visual
103	Insulator plug improperly seated		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279862
4.4.2.8	Ignition Assembly				NEXT HIGHER ASSEMBLY 9279861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Charge holder assembly		100%	3.7.1	Test(Gage)
Minor 201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279869
4.4.2.9	Cup				NEXT HIGHER ASSEMBLY 9279861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Evidence of tin plating missing		0.40%	3.2	Visual
102	Inside diameter after tin plating, max.		0.40%	3.2	Gage
103	Total length, min.		0.40%	3.2	Gage
104	Wall thickness, min.		0.40%	3.2	Gage
105	Base thickness, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279861
4.4.2.10	Detonator, Electric, Patriot Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup cracked or split		100%	3.2	Visual
Major					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Length, max.		0.40%	3.2	Gage
103	Excessive solder		0.40%	3.2	Visual
104	Evidence of flux after cleaning		0.40%	3.10	Visual
105	Solder not continuous		0.40%	3.10	Visual
106	Evidence of cold solder connections		0.40%	3.10	Visual
107	Insulation resistance		100%	3.8.2	4.5.5
108	Bridge resistance of detonator assembly		100%	3.8.1	4.5.4
109	Seal integrity		100%	3.8.7	4.5.10
110	Radiographic examination		100%	3.8.8	Test
111	No fire test	20	Note a	3.8.3	4.5.6
112	Static discharge	20	Note a	3.8.4	4.5.7
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES: (a) Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.11	Environmental and Functioning Tests for Detonator, Electric, Patriot Assembly		9279861
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inspection Lot Size 0 - 2500 / Sample Size (Groups I and II) 175				
	Group I at 70°F ± 10°F (Note a)	150	(Note c)	3.9.2	
Critical					
1	Functioning during T.V.			3.9.3	Test
2	Functioning during F.V.			3.9.4	Test
Major					
101	Transportation Vibration			3.9.3	Test
102	Flight Vibration			3.9.4	Test
103	Insulation Resistance			3.8.2	4.5.5
104	Seal Integrity			3.8.7	4.5.10
	Group I-A at 70°F ± 10°F (Note b)	50	(Note c)	3.9.2	4.5.11
105	Bridge Resistance			3.8.1	4.5.4
106	Function Time and Output			3.8.5	4.5.8
				3.8.6	4.5.9
	Group I-B at -40°F - 5°F (Note b)	50	(Note c)	3.9.3	4.5.11
107	Bridge Resistance			3.8.1	4.5.4
108	Function Time and Output			3.8.5	4.5.8

NOTES: (a) Each sample unit shall be subjected to all the tests in the group.
(b) Samples for these sub-groups shall be selected from the initial Group I test samples
(c) Accept on 0 defects, reject on 1 or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Environmental and Functioning Tests for Detonator, Electric, Patriot Assembly (Cont'd)	2 OF 2	9279861
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Group I-C at 150°F + 5°F (Note b)	50	(Note c)	3.9.2	4.5.11
109	Bridge Resistance			3.8.1	4.5.4
110	Function Time and Output			3.8.6	4.5.9
	Group II (Note a)	25	(Note c)		
Critical					
3	Functioning during T.H.			3.9.5	Test
Major					
111	Temperature and Humidity, 14 Days			3.9.1, 3.9.5	MIL-STD-331 Test 105
112	Insulation Resistance at 70°F ± 10°F			3.8.2	4.5.5
113	Seal Integrity at 70°F ± 10°F			3.8.7	4.5.10
114	Bridge Resistance at 70°F ± 10°F			3.8.1	4.5.4
115	Function Test and Output at 70°F ± 10°F			3.8.5	4.5.8
				3.8.6	4.5.9

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Box, Prior to Sealing				See Section 5
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of detonators		0.40%	3.2	Visual
102	Detonators in bundle not completely encased in foil		0.40%	3.2	Visual
103	Packet assembly torn or insecurely closed (when applicable)		0.40%	3.2	Visual
104	Package marking missing, incorrect or illegible		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Box, Sealed				See Section 5
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged to the extent that contents are exposed or liable to become exposed		0.40%	3.10	Visual
102	Marking missing, incorrect or illegible		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-48175B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9279862 and 9279861
4.4.2.14	Moisture Content of Explosives				NEXT HIGHER ASSEMBLY 9279861
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Moisture Content, Lead Azide		see 4.4.3.1	3.4.1	4.5.1
102	Moisture Content, PETN		see 4.4.3.1	3.4.2	4.5.1

NOTES:

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4.4.3 Testing.

4.4.3.1 Moisture content. This test shall be performed on a two (2) gram sample from each batch of explosive material prior to start of loading operations, and such additional tests as are warrented by variations in atmospheric conditions, process and materials. The contractor shall provide adequate controls to insure compliance with the requirements of 3.4.1 and 3.4.2 and shall test for verification one (1) sample of each material from each eight (8) hours production of detonators. Samples from composites shall not be used. If the moisture content of a sample exceeds the requirements, that quantity of material that is represented by the sample, together with all detontors loaded with the nonconforming material, shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

4.5 Methods of inspection.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Moisture content test.

a. Place a two (2) gram sample from the batch in a dry, tared container immediately prior to loading. If a delay is encountered between sampling and testing, the sampling container used shall have approximately the same volume as the sample and shall be equipped with an air tight cover.

b. Remove cover and dry in an oven at 50 $\pm$ 2°C and absolute pressure of 50 mm Hg maximum from 30 minutes.

c. Replace cover and place in a desiccator until cool.

d. Dry sample until successive weighings at 30 minute intervals agree to within $\pm$ 0.002 grams (gm).

e. Weigh to an accuracy of $\pm$ 0.2 milligrams.

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f. Calculate the moisture content as the percentage of the sample weight.

$$\text{Percent Moisture and Volatiles} = \frac{A \times 100}{B}$$

Where:

A = Basic sample weight, gm.
B = Weight of sample, gm.

4.5.1.1 Alternative. Test shall be performed in accordance with Method 101.6 of MIL-STD-650.

4.5.2 Plug insulation resistance. A potential of 1000 volts (R.M.S.) shall be applied between: (1) lead wires, and (2) the shunted lead wires and the eyelet (two separate measurements) for at least one second. The test temperature and relative humidity shall be $70^{\circ} \pm 10^{\circ}$F and not less than 80 percent respectively.

4.5.3 Plug pressure test. The unbridged plug shall be tested in a compressive test fixture such as that shown in Figure 2 in such a manner that a minimum dead load of 300 pounds applied normal to the flush surface of the plug is transmitted directly to the interface between the metal eyelet and the glass.

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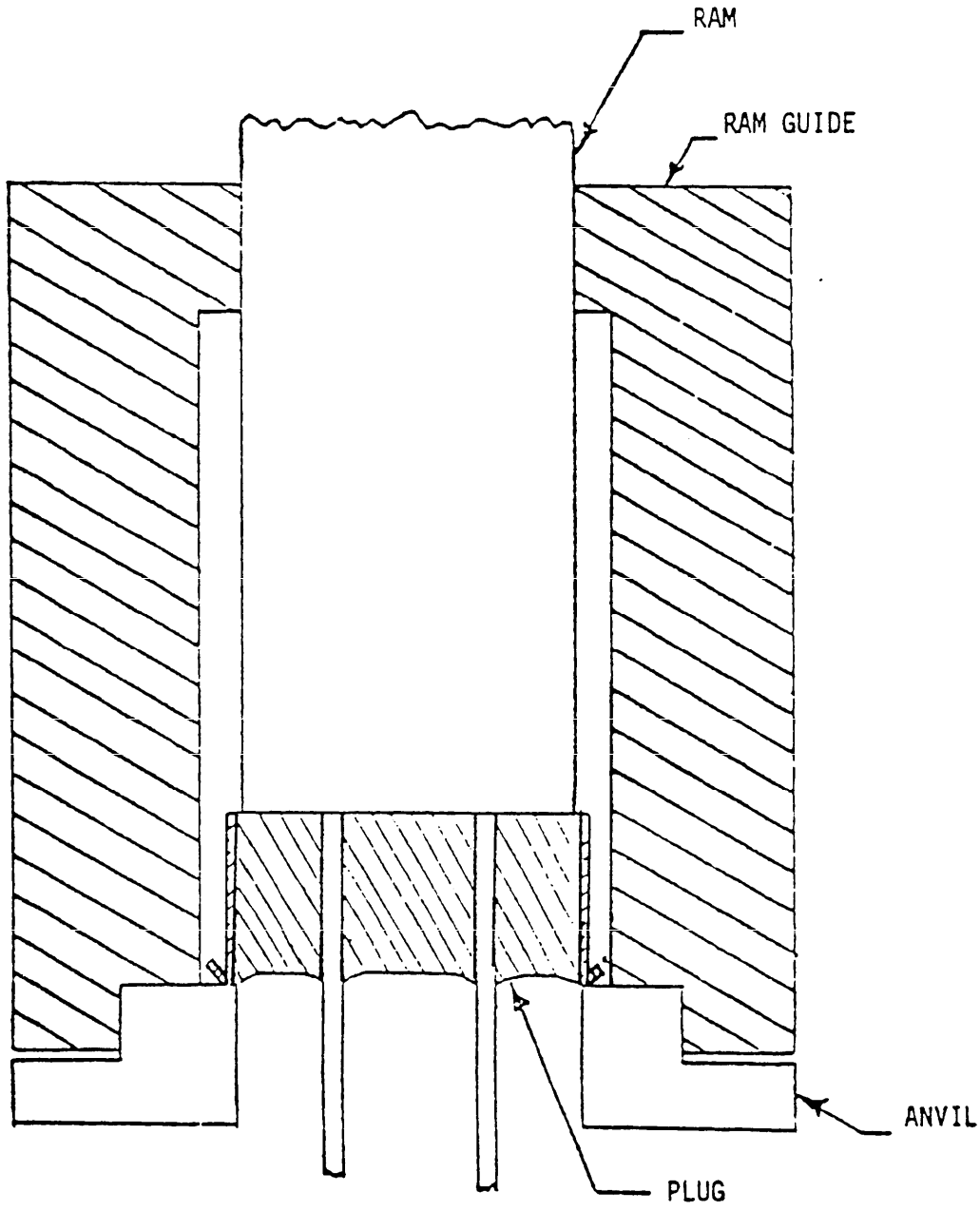


Figure 2

Schematic for Plug Pressure Test Fixture

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Improper assembly Visual

4.5.4 Bridge resistance. Caution must be observed to limit the current of the test equipment to ten (10) milliamperes and is accurate to ± 2 percent within the range of 0.5 to 2.0 ohms.

WARNING: This test shall be performed behind a suitable barricade.

4.5.5 Insulation resistance. A minimum potential of 500 volts DC shall be applied between the shunted lead wires and the cup for at least one (1) second. The test temperature and relative humidity shall be 70° ± 10°F and not less than 80 percent respectively unless otherwise specified.

WARNING: This test shall be performed behind a suitable barricade.

4.5.6 No fire. The bridge element circuit of the detonator shall be subjected to a direct current of not less than one (1) ampere for a period of five (5) minutes minimum or until actuation of the detonator, whichever is the shorter period. To insure constant direct current, the test circuit shall have resistance not less than 50 times the total resistance of the bridge circuit. For the purpose of adjusting the test current to the desired value, a resistor shall be connected in place of the detonator and the test circuit resistance or potential shall be within 10 percent of the total resistance of the bridge circuit. Subsequently, the detonator shall be connected in place of the resistor and the test current applied. There should be no circulation of air in the safety chamber in which the detonator is placed. If a rectified current is used, the RMS ripple current shall not exceed five (5) percent of the test current. Detonators used for this test shall be temperature conditioned for a minimum time of four (4) hours at the required temperature and subsequently tested under the same conditions.

WARNING: This test shall be performed behind a suitable barricade.

4.5.7 Static discharge test. The detonator shall be subjected to the discharge of 500 picofarad ± 5 percent capacitor charged to 25,000 ± 500 volts with a 5000 ohms ± 5 percent resistor in series with test potential, the shunted bridge wire circuit and the case.

4.5.8 Functioning time. The PATRIOT detonators shall be assembled as shown in Figure 3. The detonator lead wires shall be connected to the test circuit which discharges (see Figure 1) capacitors charged to 80.0 volts minimum. The functioning time shall be determined by the time of the first interruption of the capacitor discharge curve of a circuit that monitors the voltage across the bridge element of the detonator, or an equivalent test procedure that measures and records the time interval between the start of output capacitor discharge input to the detonator and its explosion.

CAUTION: The detonator shall be placed in a suitable chamber to perform this test. If the detonator fails to explode, wait a minimum of 10 minutes before removing the detonator from the safety chamber.

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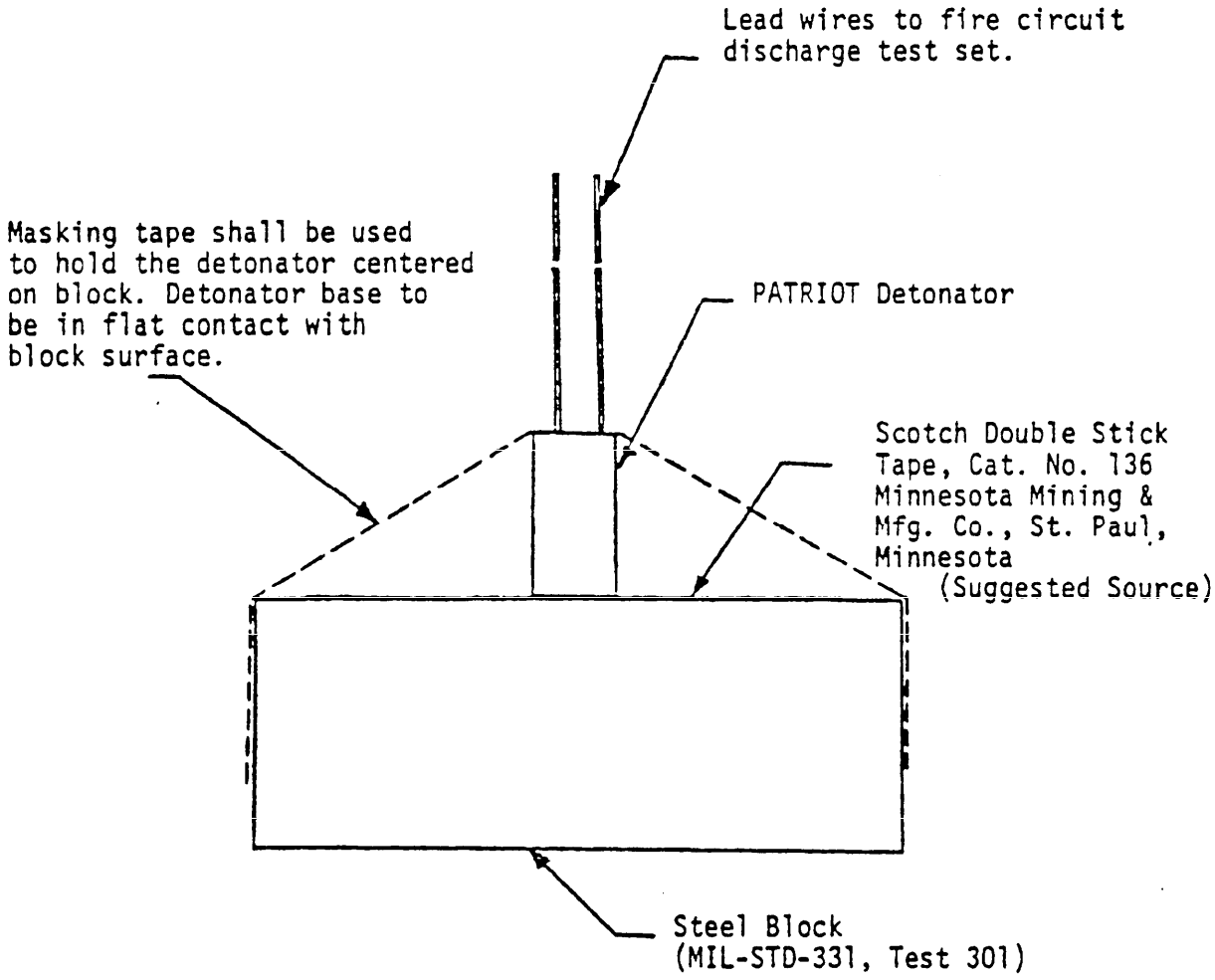


Figure 3

Steel Indentation Assembly for Patriot Detonator

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4.5.9 Output. The output of the detonator shall be determined by indentation in the steel block as specified in MIL-STD-331 Test 301.1 except no sleeve is required. The point of the dial indicator probe shall have a $30^{\circ} \pm 1^{\circ}$ included angle and the end of the point shall have a radius of 0.025 ± 0.002 inch. Detonators which fail to explode shall be classified defective in the function test requirement only, and an additional unit from the lot shall be selected to test for output requirements.

CAUTION: The detonator shall be placed in a suitable chamber to perform this test. If the detonator fails to explode, wait a minimum of 10 minutes before removing the detonator from the safety chamber.

4.5.10 Seal integrity. Each detonator shall be coated with liquid leak detector (see 6.5) which shall indicate leakage by the formulation of foam when the detonator is subjected to a vacuum with a maximum absolute pressure of 4.0 inches of Hg. A bell jar or other chamber such that the entire soldered joint may readily be observed when the required vacuum has been obtained. The detonator shall remain in the required vacuum for a minimum of one minute prior to observing for foam. Upon completion of testing the liquid leak detector shall be removed by approved cleaning solution.

4.5.11 Temperature conditioning. For tests involving extreme temperature conditions, the detonator shall be temperature conditioned for a minimum of two (2) hours after the chamber temperature has been stabilized. The detonators shall be tested in the chamber.

4.5.12 Tolerances.

4.5.12.1 Environmental (mechanical). Tolerances shall be those specified in MIL-STD-810.

4.5.12.2 Measurements (electronic). Tolerances specified herein are absolute with no allowance made for measuring equipment inaccuracy. Measurements shall ensure that the values measured are within the tolerances or limits specified.

5. PACKAGING

5.1 Packing and marking.

5.1.1 Preservation. Level C. A bundle of 25 detonators shall be made, lead wires out straight. Solid tin plated copper wire 24 AWG shall be wrapped around the lead wires to secure the bundle. Each bundle of detonators shall be wrapped in .0015 inch thick aluminum foil complying with MIL-A-148, so that all detonators and lead wires in a bundle are completely encased by the foil. Each package will be sealed in a moisture vapor barrier packaging material (see 6.5). Eight (8) of these bags shall be sealed in a large moisture vapor barrier packaging material and then packaged in a wooden box containing appropriate dunnage (sawdust).

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5.2 Packing.

5.2.1 Level C. Unit packages shall be packed in accordance with the requirements of Code of Federal Regulations CFR 49-100-190 requirements for electric blasting caps and in a manner that will insure safe delivery from the manufacturer's plant to the first receiving activity.

5.3 Marking. Each unit package shall be labeled, stamped or printed with the following information:

200 Each - DETONATOR, ELECTRIC, PATRIOT
Lot (Mfg. SYMBOL - NO.)
LOADED (Mo/Yr)
DANGEROUS - HANDLE CAREFULLY

Letters shall be upper case. Size of marking 1/4 inch minimum. Ink shall be black, waterproof commercial grade.

Labels shall be white paper of 15 pound basic weight minimum.

Exterior boxes shall be marked in accordance with dwg. 8796522. Cleat markings are not applicable when fiber boxes are used. DOT markings shall be:

*(QTY) - ELECTRIC BLASTING CAPS - HANDLE CAREFULLY

*Insert number of detonators in the exterior container.

5.4 Data cards. Data cards shall be prepared for each lot (see 6.2 and Table II).

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TABLE II

COMPONENTS LIST FOR AMMUNITION DATA CARDS

COMPONENT	DRAWING NUMBER	MODEL	MANUFACTURED	DATE MANUFACTURED	LOT NO.	QUANTITY
Bridge Element	9279867	NA	Required	Required	If Available	Required
Eyelet	9279866	NA	"	"	"	"
Cup	9279869	NA	"	"	"	"
Explosives						
Lead Azide	9279862	NA	"	"	"	"
PETN	9279861	NA	"	"	"	"

NA - Not Applicable

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6. NOTES

6.1 Intended use. The Detonator, Electric, PATRIOT is intended for use in the Safety and Arming Device: M143 (M143E1) for the PATRIOT Surface to Air Missile.

6.2 Ordering data. See 6.1 of MIL-A-48078.

6.2.1 Contract data requirements. The requirements for submission of data from the No Fire Test, Static Discharge Test and Environmental Test of 4.3.4.1 and 4.4.2.11 will be listed directly on a DD Form 1423 incorporated into the contract. Data Item Description DI-T-1906 will be required in connection with this Specification.

6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-I, Dover, New Jersey 07801-5001.

6.4 Liquid leak detector. A suggested source is:

American Gas and Chemical Co., Inc.
Department R&D
120 East End Avenue
New York, New York

"Leak Tec" No. 577-V

6.5 Moisture vapor barrier packaging material. A suggested source is:

Customs Material Inc.
Chelmsford, Massachusetts

"VELO-PAK-1701 VP" Bag

6.6 Check test for possible deterioration. The following test is designed to prevent detonators that have deteriorated being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than five (5) years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of that check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Two hundred fifteen (215) detonator assemblies shall be selected from each lot and tested as specified in 4.4.3.

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6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. ARRADCOM technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.8 Key word listing.

Detonator (S&A)

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1336-A532)

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(See Instructions – Reverse Side)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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